Semester: 6th
Subject Name:- SPM & Code:-CS30036
Branch (s): - CSE/IT

SPRING MID SEMESTER EXAMINATION-2026

School of Computer Engineering
Kalinga Institute of Industrial Technology, Deemed to be University
Subject Name: Software Project Management
[Subject Code: CS30036]

Time: 1 1/2 Hours **Full Mark: 20**

Answer any four questions including question No.1 which is compulsory.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

1. Answer all the questions. [1 Mark X 5]

a) The size of a project is 600 FP. Using Caper Jones' rules, estimate effort.
b) In a Mark II Function Point analysis, if number of input data elements is 30, number of output data elements is 20 and number of entities accessed is 10, calculate the UFP.
c) If a project's estimated effort is 100 person-months and development time is 10 months, calculate the average team size.
d) If Actual Cost(AC)= 15 days and Cost Variance(CV)= 5 days and Budget at Completion (BAC) = 1000$. Calculate Estimate at Completion(EAC)?
e) The initial investment and year-wise cash flow for four years on a project is given below.

Year	Project cash flow
0	-200,000
1	50,000
2	75,000
3	100,000
4	150,000

Calculate the pay back period.

2. Two competing software projects A and B have the following expected cash flows. Assuming a discount rate of 20%, recommend which project should be selected and justify your answer.

Project A		Project B	
Year	Cash Flow (₹ lakhs)	Year	Cash Flow (₹ lakhs)
0	–20	0	–30
1	+10	1	+17
2	+12	2	+13
3	+15	3	+10

[5 Marks]

3. A software project is estimated to have 60000 LOC. If the project uses the COCOMO II Post-Architecture Model, then estimate the effortand total project cost with the following parameters:

A = 2.94;

Scale factors' values are PREC=4.96, FLEX=1.01, RESL=4.24, TEAM=3.29 and PMAT=3.12;

Effort multipliers' values are required_re-usability(RESU) = 1.24, personal_experience (PREX) = 0.62, schedule_pressure (SCED) = 1.43 and rest of the factors are nominal.

Cost per person-month = ₹80,000. [5 Marks]

4. Compare step-wise planning with traditional single-stage planning in project management. Highlight the advantages and disadvantages. [5 Marks]

5. Why Configuration Control Management is necessary? Explain the configuration control process with a suitable block diagram. [2.5+2.5 Marks]

*** Best of Luck ***